

PROFESSIONAL PROFILE

Dedicated and aspiring biotechnologist with a strong academic background in molecular biology, genetics, and biochemistry. Passionate about advancing biotechnological innovations to improve healthcare, agriculture, and environmental sustainability. Experienced in performing data analysis, and documenting results. Adept at collaborating in multidisciplinary teams, adapting to new challenges, and committed to continuous learning and professional development. Passionate about continuous learning and staying updated with the latest technologies in Unix, shell scripting, Python development, SQL, AWS, and DevOps.

IT TECHNICAL SKILLS

- **Cloud Computing:** AWS
- **Data Warehouse:** Snowflake
- **Programming Languages:** Python
- **Databases:** DynamoDB, MySQL
- **Scripting:** Shell Scripting, Python
- **Infrastructure as Code:** Terraform
- **AWS Services:** EMR EC2, S3, IAM, CloudFormation, SNS, CloudWatch, RDS, ECS
- **Version Control:** GitHub
- **CI/CD:** Jenkins
- **Containers:** Docker
- **Project Management Tools:** Jira

BIOTECHNOLOGY TECHNICAL SKILLS

- Gel Electrophoresis (AGE, SDS PAGE, NATIVE PAGE)
- ELISA
- Western Blot
- Ficoll-Hypaque technique
- PCR
- RT-PCR
- Chromatography - TLC, Affinity, Ion Exchange, Gel Filtration, Column
- Plasmid DNA Isolation
- DNA/ RNA Extraction
- Cloning
- Transformation
- Transfection
- Animal Cell Culture
- FACS
- Invivo Handling (Mice) – Administration, Sacrificing and Tissue collection
- Bioinformatics Tools like String, RasMol, PyMol, Sanger Sequencing, NGS

EDUCATION

- Master of Technology, Biotechnology, University College of Engineering, BIT Campus, Anna University, Chennai (9.11)
- Bachelor of Technology, Biotechnology, Vivekananda College of Engineering for Women, Anna University, Chennai (9.47)

PROFESSIONAL EXPERIENCE

LANGUAGE INTERPRETER | LANGUAGELINE SOLUTIONS | Feb 2025 – June 2025

Responsibilities:

- Provide real-time, accurate, and culturally appropriate interpretation services between English and Tamil for clients across various industries including healthcare, legal, and customer service.
- Maintain strict confidentiality and adhere to professional codes of ethics for interpreters.
- Handle a high volume of calls efficiently while ensuring clarity and comprehension for all parties.
- Interpret complex and sensitive information with precision, maintaining the tone and intent of the original message.
- Continuously enhance linguistic and cultural knowledge to provide top-quality interpretation services.

DATA ANNOTATION SPECIALIST | CROWDGEN | Nov 2024 - June 2025

- Annotated large volumes of text/image/audio data sets used to train and improve AI/ML models with high attention to detail.
- Followed strict labeling guidelines and project-specific protocols to ensure consistency and accuracy.
- Achieved >95% quality rating / met weekly targets consistently.
- Collaborated with QA reviewers and participated in feedback loops for model improvement.

RESEARCH INTERN (JUNIOR SPECIALIST) IN UNIVERSITY OF CALIFORNIA, SAN DIEGO | JULY 2018- SEP 2018

Project Title : *Invivo* Gene therapy in a mouse model for DMD

Worked in RNA editing enzyme, ADAR2 which is used to repair mutations. We engineered Adenosine Deaminases in RNA (ADAR), which converts Adenosine to Inosine (biologically recognized as Guanosine) to target point mutation causing muscular dystrophy. Specifically, by delivering the RNA editing enzyme, ADAR2 and an associated guiding RNA (adRNA) via adeno-associated viruses, we enabled premature stop codon read-through and correction in the *mdx* mouse model of muscular dystrophy that harbors a nonsense mutation in the dystrophin gene editing. This resulted in restoration of dystrophin in *mdx* mouse model. With this result, we also tried splicing disruption in Voltage gated Sodium Channel (Nav 1.7) for pain management using ADAR. We used techniques like transformation, Plasmid DNA Isolation, Sanger sequencing, Agarose Gel Electrophoresis, Mammalian cell culture, transfection, RNA isolation and cDNA synthesis, transfection, qPCR quantification, FACS and Western blot.

ACADEMIC PROJECT EXPERIENCE

UG PROJECT

PROJECT 1

Title: Medium Optimization of medicinal mushrooms for the treatment of oral Cancer

PROJECT 2

Title: Formulation of principle compounds from *Areca catechu* (Betelnut)

PROJECT 3

Title: Implementation of antagonistic microorganisms for the treatment of Turmeric (*Curcuma longa* L.,) rhizome rot

PG PROJECT

Title: Prevention Of Hyperglycemia Induced β -cell Death by Silencing ER Stress Response Using Artificial MiRNA

Worked in RNA interference (RNAi), a powerful tool for the regulation of gene expression that regulate gene expression and protein coding by base pairing with the 3' untranslated region (UTR) of target mRNAs to treat diabetes. We have successfully cloned amiRNAs in the pcDNA 3.1 (+) vector which helped in silencing ER stress response to prevent hyperglycemia induced β -cell death by targeting CHOP, ASK1 and GADD45 α .

TATA CONSULTANCY SERVICES LTD | ASSISTANT SYSTEM ENGINEER TRAINEE FEB 2013 – OCT 2013

Client – Morgan Stanley | Unix, SQL

Morgan Stanley is a global financial services firm that provides a range of investment banking, securities, wealth management, and investment management services. It operates in various financial markets and serves institutional, corporate, and individual clients.

Responsibilities

- Responsible for incident response and resolution within the SLA.
- Reporting to the client and participating in day-to-day meetings to discuss incident reports and develop permanent solutions for all production problems.
- Responsible for identifying and solving issues within multiple components of critical business systems. Typical day involves providing dynamic service to ensure the business operations are running smoothly.
- Responsible for identifying and solving issues within multiple components of critical business systems. Typical day involves providing dynamic service to ensure customer service operations are running smoothly.
- Maintain production environments.
- Perform root cause analysis for production errors.
- Investigate and resolve technical issues.
- Work with software developers and software engineers to ensure that development follows established processes and works as intended.